



Differential Equation Topic Test 1

Solution

Total Marks: 30

Time: 50 minutes

- 1 Which one of the following differential equations has $y = 5xe^{2x}$ as a solution? 1

A. $\frac{d^2y}{dx^2} - 2\frac{dy}{dx} = 0$

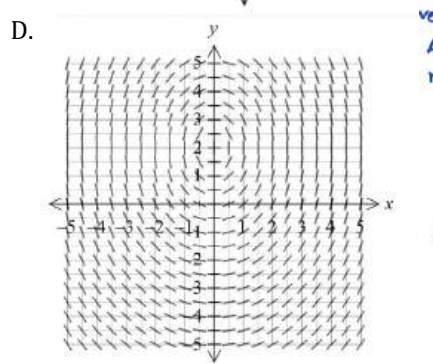
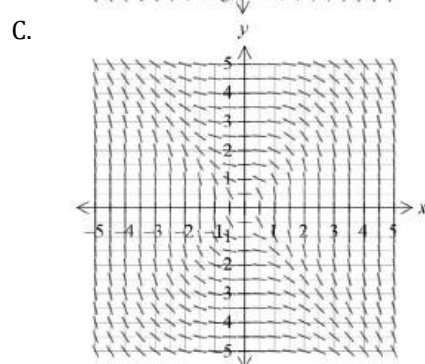
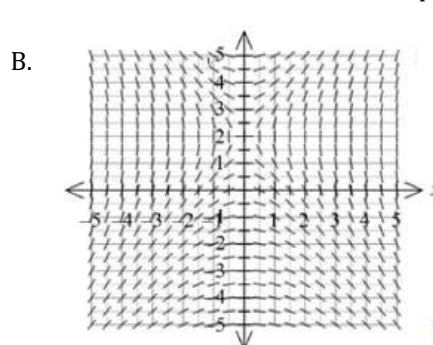
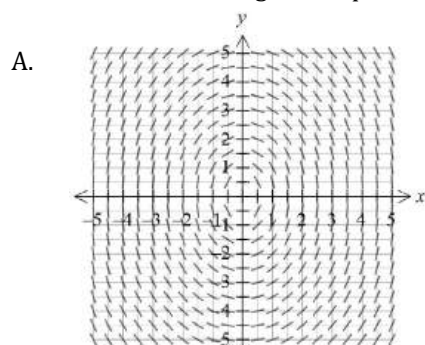
B. $\frac{d^2y}{dx^2} - 4\frac{dy}{dx} = 0$

C. $\frac{d^2y}{dx^2} - 4y = 20e^{2x}$

D. $\frac{d^2y}{dx^2} - 4y = e^{2x}$

$y = 5xe^{2x}$
 $\frac{dy}{dx} = 5e^{2x} + 10xe^{2x}$ $\begin{cases} u=5x & v=e^{2x} \\ u'=5 & v'=2e^{2x} \end{cases}$ C
 $\frac{d^2y}{dx^2} = 10e^{2x} + 10e^{2x} + 20xe^{2x}$ $\begin{cases} u=10x & v=e^{2x} \\ u'=10 & v'=2e^{2x} \end{cases}$
 $\frac{d^2y}{dx^2} = 20e^{2x} + 20xe^{2x}$
 $\textcircled{C} \quad \frac{d^2y}{dx^2} - 4y = 20e^{2x}$
 $\text{LHS} = 20e^{2x} + 20xe^{2x} - 4(5xe^{2x})$
 $= 20e^{2x} + 20xe^{2x} - 20xe^{2x}$
 $= 20e^{2x}$
 $= \text{RHS}$

- 2 Which of the following best represents the direction field for the differential equation $\frac{dy}{dx} = -\frac{2x}{y-2}$? 1



As $y=2$ makes denominator zero, vertical gradient for points with $y=2$
 As $x=0$ makes numerator zero horizontal gradient for points with $x=0$
 This only happens at B and D
 By substituting coordinate values into $\frac{dy}{dx}$ option D is correct.
 e.g. Test $(3,0) \quad \frac{dy}{dx} = 3$

- 3 Find the solution of the differential equation $\frac{dy}{dx} = 1 + 4y^2$, given $y(0) = \frac{1}{2}$. Give your answer in the form 3

$y = f(x).$
 $\frac{dy}{dx} = 1 + 4y^2$
 Separating Variables
 $\frac{1}{1+4y^2} dy = dx$
 Integrating both sides
 $\int \frac{1}{1+(2y)^2} dy = \int dx$
 $\frac{1}{2} \tan^{-1} 2y = x + C$
 Sub $y(0) = \frac{1}{2}$
 $\frac{1}{2} \tan^{-1} 2(\frac{1}{2}) = 0 + C$
 $C = \frac{1}{2} (\frac{\pi}{4}) = \frac{\pi}{8}$
 $\therefore \frac{1}{2} \tan^{-1} 2y = x + \frac{\pi}{8}$
 $\tan^{-1} 2y = 2x + \frac{\pi}{4}$
 $2y = \tan(2x + \frac{\pi}{4})$
 $y = \frac{1}{2} \tan(2x + \frac{\pi}{4})$

- 4 The logistic equation $\frac{dP}{dt} = a \left(1 - \frac{P}{k}\right) P$ can be used to model the size of a population P , over time t . If a and k are both positive constants and $0 < P < k$, which of the following statements must be true?

1

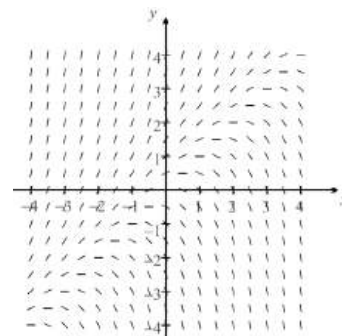
- A. the population is increasing
B. the population is constant
C. the population is decreasing
D. none of the above

$$\begin{aligned} & \text{If } P < k \\ & 0 < \frac{P}{k} < 1 \\ & 1 - \frac{P}{k} > 0 \quad \therefore (A) \\ & \therefore \frac{dP}{dt} > 0 \\ & \therefore \text{population is increasing} \end{aligned}$$

- 5 Which of the following differential equations could produce the slope field shown above?

1

- A. $\frac{dy}{dx} = y^2 - 1$
B. $\frac{dy}{dx} = x + y$
C. $\frac{dy}{dx} = y - x$
D. $\frac{dy}{dx} = y - 1$



$$\begin{aligned} & \bullet \text{ Consider } \frac{dy}{dx} = y^2 - 1. \text{ at } y=0, \frac{dy}{dx} = -1 \\ & \text{but on slope field } \frac{dy}{dx} \neq -1 \text{ at } y=0. \\ & \therefore \text{Not (A).} \\ & \bullet \text{ Consider } \frac{dy}{dx} = x + y \text{ at } (-4, -4). \\ & \frac{dy}{dx} = -8 \text{ on slope field } \therefore \text{Not (B).} \\ & \bullet \text{ Consider } \frac{dy}{dx} = y - x \text{ at } (-4, -4). \\ & \frac{dy}{dx} = 0, \text{ agrees with slope field.} \\ & \text{at } y=0, \frac{dy}{dx} = -x. \text{ Agrees with slope field.} \\ & \text{at } x=0, \frac{dy}{dx} = y. \text{ Also agrees.} \\ & \bullet \text{ Check } \frac{dy}{dx} = y - 1 \text{ at } (-4, -4). \\ & \frac{dy}{dx} = -5 \therefore \text{Not (D)} \therefore (C) \end{aligned}$$

- 6 Solve the differential equation $\frac{dy}{dx} = 3x^2 e^{-y}$, given $y(0) = 1$. Express your answer with y as the subject.

3

$$\begin{aligned} & \frac{dy}{dx} = 3x^2 e^{-y} \quad y(0) = 1 \\ & e^y dy = 3x^2 dx \\ & \int e^y dy = \int 3x^2 dx \quad \checkmark \\ & e^y = x^3 + c \\ & y = \ln(x^3 + c) \quad \checkmark \\ & y(0) = 1 \\ & 1 = \ln(c) \\ & c = e \\ & y = \ln(x^3 + e) \quad \checkmark \end{aligned}$$

- 7 (i) Let $y = xu$. Given that u is a function of x , find an expression for $\frac{dy}{dx}$.

1

$$\begin{aligned} & \text{i) } y = xu \quad \text{if } y = uv \quad y' = vu' + uv' \\ & \frac{dy}{dx} = u + x \frac{du}{dx} \quad \checkmark \end{aligned}$$

- ii) Hence solve the initial value problem $\frac{dy}{dx} = \frac{y^2}{2x^2} + \frac{1}{2}$, given $y(1) = 2$. Express your answer with y as the subject.

2

$$\begin{aligned}
 & \text{ii) } y' = \frac{y^2}{2x^2} + \frac{1}{2} \quad y(1) = 2 \\
 & \text{Since } y' = u + xu' \text{ and } y = xu \\
 & u + xu' = \frac{(xu)^2}{2x^2} + \frac{1}{2} \\
 & u + xu' = \frac{u^2}{2} + \frac{1}{2} \\
 & xu' = \frac{u^2}{2} + \frac{1}{2} - u \\
 & xu' = \frac{u^2 - 2u + 1}{2} \\
 & xu' = \frac{1}{2}(u-1)^2 \\
 & x \frac{du}{dx} = \frac{1}{2}(u-1)^2 \\
 & \frac{1}{(u-1)^2} \frac{du}{dx} = \frac{1}{2x} \\
 & \int \frac{1}{(u-1)^2} du = \int \frac{1}{2x} dx \\
 & \int (u-1)^{-2} du = \int \frac{1}{2x} dx \\
 & \frac{(u-1)^{-1}}{-1} = \frac{1}{2} \ln|x| + C \\
 & -\frac{1}{u-1} = \frac{1}{2} \ln|x| + C \\
 & \frac{1}{1-u} = \frac{1}{2} \ln|x| + C \\
 & \frac{1}{1-\frac{y}{x}} = \frac{1}{2} \ln|x| + C \\
 & 1 - \frac{y}{x} = \frac{1}{\frac{1}{2} \ln|x| + C} \\
 & 1 - \frac{1}{\frac{1}{2} \ln|x| + C} = \frac{y}{x} \\
 & y = x - \frac{x}{\frac{1}{2} \ln|x| + C} \\
 & y = x - \frac{x}{\ln|x| + C}
 \end{aligned}$$

- 8 (i) Show that $\sin^3 x = \sin x - \cos^2 x \sin x$.

1

$$\begin{aligned}
 \text{LHS} &= \sin^3 x \\
 &= \sin x (1 - \cos^2 x) \\
 &= \sin x - \cos^2 x \sin x
 \end{aligned}$$

- (ii) Use the substitution $u = \cos x$ to find $\int \cos^2 x \sin x dx$.

1

$$\begin{aligned}
 u &= \cos x \quad \frac{du}{dx} = -\sin x \\
 \int \cos^2 x \sin x dx &= -\int u^2 \frac{du}{dx} dx \\
 &= -\frac{u^3}{3} + C = -\frac{\cos^3 x}{3} + C
 \end{aligned}$$

- (iii) Hence find the general solution to the differential equation $\operatorname{cosec}^3 x \frac{dy}{dx} = \cos^2 y$. You may

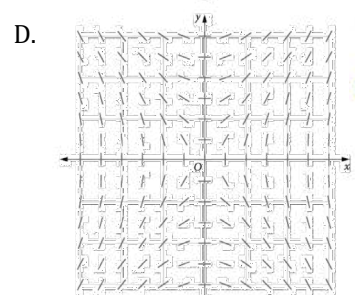
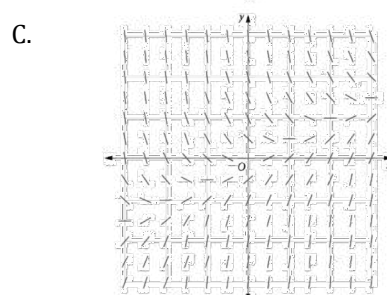
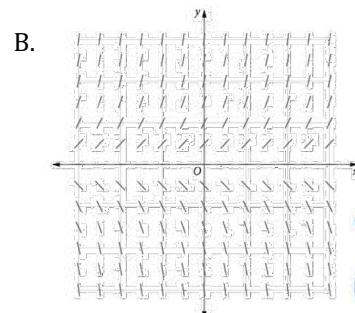
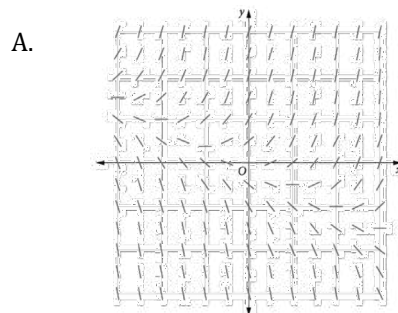
assume $-\frac{\pi}{2} < y < \frac{\pi}{2}$.

3

$$\begin{aligned}
 \operatorname{cosec}^3 x \frac{dy}{dx} &= \cos^2 y \\
 \int \frac{1}{\cos^2 y} dy &= \int \frac{1}{\operatorname{cosec}^3 x} dx \\
 \int \sec^2 y dy &= \int \sin^3 x dx \\
 \tan y + A &= \int \sin^2 x \cdot \sin x dx \quad (\text{for } \tan y) \\
 \tan y + A &= \int (1 - \cos^2 x) \sin x dx \\
 \tan y + A &= \int \sin x - \cos^2 x \sin x dx \\
 \tan y + A &= -\cos x - \left[-\frac{\cos^3 x}{3} \right] + B \\
 \tan y &= -\cos x + \frac{\cos^3 x}{3} + C \\
 y &= \tan^{-1} \left(-\cos x + \frac{\cos^3 x}{3} + C \right)
 \end{aligned}$$

- 9 Which of the following could have the differential equation $\frac{dy}{dx} = x - ky$ as a solution if $k > 0$?

1



when $\frac{dy}{dx} = 0$: $ky = x \Rightarrow y = \frac{x}{k}$
 (line through origin, having horizontal line elements.
 $\therefore$ not (B) and not (D).
 But $k > 0$, $\therefore y = \frac{x}{k} \rightarrow \therefore$ (C)

- 10 A saltwater fish tank contains 250 litres of liquid. To keep the water clean, a pump draws out 15 litres per minute. To keep the amount of liquid in the tank constant, 15 litres of a saline solution with a salinity of 32.4 grams per litre is pumped into the tank every minute. Assume that the salt dissolves evenly into the solution in the tank.



Let $Q(t)$ be the amount of dissolved salt in the fish tank after t minutes. The tank initially contains 8070 grams of salt dissolved in the liquid.

(i) Show that $\frac{dQ}{dt} = \frac{24300 - 3Q}{50}$.

- (ii) Hence solve the differential equation for the solution to the amount of salt $Q(t)$ in the tank at any time.

i. Rate of inflow = 486 g/min
 Rate of outflow = $\frac{Q}{250} \times 15$ g/min

$$\frac{dQ}{dt} = 486 - \frac{15Q}{250}$$

$$= \frac{24300}{50} - \frac{3Q}{50}$$

$$\therefore \frac{dQ}{dt} = \frac{24300 - 3Q}{50}$$

ii.
$$\int \frac{1}{24300 - 3Q} dQ = \int \frac{1}{50} dt$$

$$-\frac{1}{3} \ln(24300 - 3Q) + c_1 = \frac{1}{50}t + c_2$$

$$\ln(24300 - 3Q) = -\frac{3}{50}t + c$$

$$24300 - 3Q = e^{-\frac{3}{50}t + c}$$

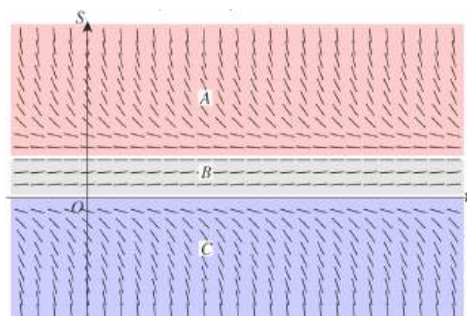
$$-3Q = Ae^{-\frac{3}{50}t} - 24300$$

$$Q = Be^{-\frac{3}{50}t} + 8100$$
 where B is some constant
 At $t = 0$, $Q = 8070$:
 $8070 = B + 8100$
 $B = -30$ $\therefore Q(t) = 8100 - 30e^{-\frac{3}{50}t}$

- 11 The spread of a flu through a population is modelled by the equation $S = \frac{2000}{1 + 199e^{-0.4t}}$ where S is the total number of people infected after t days.

(i) Show that the given equation for S satisfies the differential equation $\frac{dS}{dt} = \frac{S}{5} \left(2 - \frac{S}{1000} \right)$.

- (ii) The slope field of $\frac{dS}{dt} = \frac{S}{5} \left(2 - \frac{S}{1000} \right)$ is shown. The three regions A, B and C in which a solution curve can be found, have been labelled and shaded correspondingly. State which region the given solution curve S can exist in and justify your answer with reference to its constant solutions.



- (iii) Using this model, find on which day the rate of increase reaches a maximum. Give your answer correct to the nearest day.

i.
$$\frac{dS}{dt} = \frac{-2000(-0.4 \times 199e^{-0.4t})}{(1 + 199e^{-0.4t})^2}$$

$$= \frac{2000}{1 + 199e^{-0.4t}} \left(\frac{2}{5} \times \frac{199e^{-0.4t}}{1 + 199e^{-0.4t}} \right)$$

$$= \frac{S}{5} \times \frac{2(1 + 199e^{-0.4t} - 1)}{1 + 199e^{-0.4t}}$$

$$= \frac{S}{5} \times \frac{2(1 + 199e^{-0.4t}) - 2}{1 + 199e^{-0.4t}}$$

$$= \frac{S}{5} \left(2 - \frac{1}{1000} \times \frac{2000}{1 + 199e^{-0.4t}} \right)$$

$$\therefore \frac{dS}{dt} = \frac{S}{5} \left(2 - \frac{S}{1000} \right)$$

ii. From part (i), when $\frac{dS}{dt} = 0$:

$$\frac{S}{5} \left(2 - \frac{S}{1000} \right) = 0$$

$$S = 0 \text{ or } S = 2000$$
 $\therefore$ The logistic curve S is bounded by the two constant solutions $S = 0$ and $S = 2000$.

When $t = 0$: $S = \frac{2000}{1 + 199e^0} = 10$

The initial value $S = 10$ is within the interval $[0, 2000]$.

Hence, the solution curve S is in region B.

- iii. The graph of $\frac{dS}{dt}$ is a concave down parabola with roots $S = 0$ and $S = 2000$.

$$S_{\text{vertex}} = \frac{0 + 2000}{2} = 1000$$

When $S = 1000$:

$$1000 = \frac{2000}{1 + 199e^{-0.4t}}$$

$$1 + 199e^{-0.4t} = 2$$

$$e^{-0.4t} = \frac{1}{199}$$

$$t = -\frac{5}{2} \ln \left(\frac{1}{199} \right) = 13.233\dots$$

Hence the rate of increase is a maximum on the 14th day.

Differential Equations Topic Test 2 Solution

Total Marks: 30

Time: 50 minutes

- 1 Solve the differential equation $\frac{dy}{dx} = -\frac{x}{1+y^2}$, given that $y(-1) = 1$. Express your answer in the form 2

$ay^3 + bx^2 + cy + d = 0$, where a, b, c, d are integers.

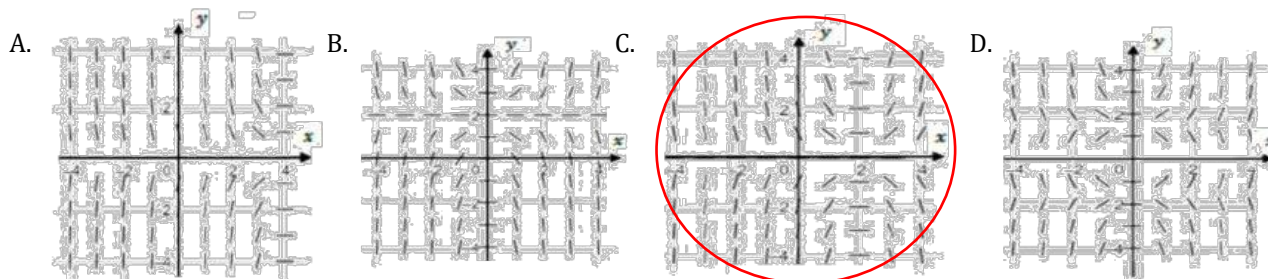
$$\frac{dy}{dx} = -\frac{x}{1+y^2} \quad dy(1+y^2) = -x dx$$

$$y + \frac{y^3}{3} = -\frac{x^2}{2} + C \text{ and } y(-1) = 1 \quad \checkmark$$

$$1 + \frac{1^3}{3} = -\frac{(-1)^2}{2} + C \rightarrow C = \frac{11}{6} \quad \checkmark$$

$$y + \frac{y^3}{3} = -\frac{x^2}{2} + \frac{11}{6} \rightarrow 2y^3 + 3x^2 + 6y - 11 = 0 \quad \checkmark$$

- 2 Which of the following best represents the slope field for differential equation $y' = xy - 2y$? 1



- 3 Find the solution of the differential equation $y' = 1 + 4y^2$ given $y(0) = \frac{1}{2}$. 2

$$\frac{dy}{dx} = 1 + 4y^2 \quad \tan^{-1} \left(2 \times \frac{1}{2} \right) = 0 + c$$

$$\int \frac{1}{1 + 4y^2} dy = \int 1 dx \quad c = \frac{\pi}{4}$$

$$\tan^{-1}(2y) = x + c \quad 2y = \tan \left(x + \frac{\pi}{4} \right)$$

$$x = 0 \Rightarrow y = \frac{1}{2} \quad y = \frac{1}{2} \tan \left(x + \frac{\pi}{4} \right)$$

- 4 A population of monkeys on a small island is an example of logistic growth. The population of the monkeys P is given by $P = \frac{50}{1+24e^{-2t}}$ and t is in years.

- (i) Determine the carrying capacity of the population.

As $t \rightarrow \infty, P \rightarrow \frac{50}{1+0}$ so the carrying capacity is 50 monkeys.

- (ii) Show that $\frac{dP}{dt} = \frac{1}{25}P(50 - P)$.

$$P = \frac{50}{1+24e^{-2t}} = 50(1+24e^{-2t})^{-1}$$

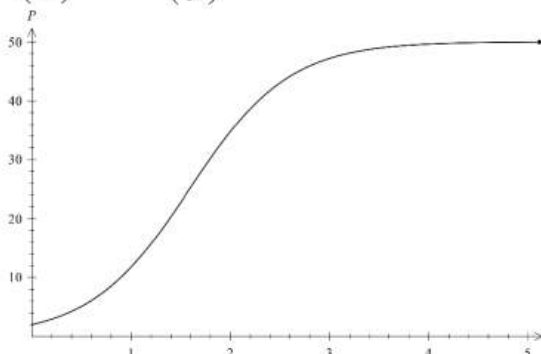
$$\frac{dP}{dt} = -50(1+24e^{-2t})^{-2} \times -48e^{-2t} \quad \checkmark$$

$$= \frac{2400e^{-2t}}{(1+24e^{-2t})^2} \text{ and } P = \frac{50}{1+24e^{-2t}} \rightarrow 1+24e^{-2t} = \frac{50}{P}$$

$$= \frac{24e^{-2t} \times 100}{\left(\frac{50}{P}\right)^2} = \frac{100 \times \left(\frac{50}{P} - 1\right)}{\left(\frac{50}{P}\right)^2} = \frac{1}{25}P(50 - P)$$

- (iii) Draw a neat sketch of $P = \frac{50}{1+24e^{-2t}}$.

the shape of the function at $t = 0$.



- 5 The population $P(t)$ of bacteria in a petri dish is modelled by the logistic differential equation

$$\frac{dP}{dt} = \frac{P}{6} \left(1 - \frac{P}{8000} \right) \text{ where } P(0) = P_0 \text{ and } t \text{ is the time in hours.}$$

- (i) If the initial population P_0 is 1000 bacteria, show that $P(t) = \frac{8000}{1 + 7e^{-\frac{t}{6}}}$. You may use the fact that

$$\frac{Q}{P(Q-P)} = \frac{1}{P} + \frac{1}{Q-P}.$$

- (ii) If instead the initial population P_0 were 12000 bacteria, describe what would have happened to the population as t increases.

$$\begin{aligned} \text{i. } \frac{dP}{dt} &= \frac{P}{6} \left(1 - \frac{P}{8000} \right) \\ \frac{dt}{dP} &= \frac{48000}{P(8000 - P)} \\ \int dt &= 6 \int \frac{8000}{P(8000 - P)} dP \\ \frac{1}{6} \int dt &= \int \left(\frac{1}{P} + \frac{1}{8000 - P} \right) dP \text{ using } \frac{Q}{P(Q-P)} = \frac{1}{P} + \frac{1}{Q-P} \\ \frac{1}{6} t + c &= \ln|P| - \ln|8000 - P| \\ \frac{t}{6} + c &= \ln \left| \frac{P}{8000 - P} \right| \\ \left| \frac{P}{8000 - P} \right| &= e^c e^{\frac{t}{6}} \\ \frac{P}{8000 - P} &= \pm e^c e^{\frac{t}{6}} \\ \frac{P}{8000 - P} &= A e^{\frac{t}{6}}, \text{ where } A = \pm e^c \\ \text{When } t = 0, P = 1000 : \\ \frac{1000}{8000 - 1000} &= A e^0 \\ \therefore A &= \frac{1}{7} \\ \therefore \frac{P}{8000 - P} &= \frac{1}{7} e^{\frac{t}{6}} \\ \therefore P &= \frac{8000}{1 + 7e^{-\frac{t}{6}}} \end{aligned}$$

- ii. The carrying capacity for the population is 8000. If $P(0) > 8000$, such as 12000, the population will decay and approach the asymptotic population size of 8000 as time increases.

- 6 Solve the differential equation $\frac{dy}{dx} = \frac{\cos^2 y}{e^x}$ given that $y(0) = \frac{\pi}{4}$. Express your answer in the form $y = f(x)$.

$$\begin{aligned} \frac{dy}{dx} &= \frac{\cos^2 y}{e^x} \\ \frac{dy}{\cos^2 y} &= \frac{dx}{e^x} \\ \int \sec^2 y \, dy &= \int e^{-x} dx \\ \tan y &= -e^{-x} + c \\ x = 0, y &= \frac{\pi}{4} \\ \tan\left(\frac{\pi}{4}\right) &= -e^{-(0)} + c \\ 1 &= -1 + c \\ c &= 2 \\ \therefore \tan y &= -e^{-x} + 2 \\ y &= \tan^{-1}(2 - e^{-x}) \end{aligned}$$

$\frac{dP}{dt} = \frac{3P}{2500}(2500 - P)$, where the initial population of mice is 500. Given $\frac{1}{P} + \frac{1}{2500-P} = \frac{2500}{P(2500-P)}$, solve the differential equation to find the population P of the mice at time t . Express your answer in the form

$P = \frac{A}{1 - Be^{-kt}}$ where A, B and k are integers.

$$\begin{aligned} \frac{dP}{dt} &= \frac{3P}{2500}(2500 - P) && \text{Trivial soln: } P = 0 \\ &&& P = 2500 \\ \frac{dP}{P(2500-P)} &= \frac{3}{2500} dt \\ \int \frac{2500}{P(2500-P)} dP &= \int 3 dt \\ \int \left(\frac{1}{P} + \frac{1}{2500-P} \right) dP &= 3 \int dt \\ \ln |P| - \ln |2500-P| &= 3t + C \\ \ln \left| \frac{P}{2500-P} \right| &= 3t + C \\ \left| \frac{P}{2500-P} \right| &= e^{3t+C} \\ &= e^{3t} \times e^C \\ &= Ae^{3t} && \text{where } A = e^C \\ \frac{P}{2500-P} &= Ae^{3t} && A = \pm e^C \\ \text{When } t=0 &&& \frac{500}{2000} = Ae^0 \\ P=500 &&& \frac{1}{4} = A \\ \therefore \frac{P}{2500-P} &= \frac{1}{4} e^{3t} \\ 4P &= e^{3t}(2500-P) \\ 4P &= 2500e^{3t} - Pe^{3t} \\ 4P + Pe^{3t} &= 2500e^{3t} \\ P(4 + e^{3t}) &= 2500e^{3t} \\ P &= \frac{2500e^{3t}}{4 + e^{3t}} \div e^{3t} \\ &= \frac{2500}{\frac{4}{e^{3t}} + 1} \\ &= \frac{2500}{1 + 4e^{-3t}} && B = -4 \quad [4] \\ \text{or } P &= \frac{2500}{1 - (-4)e^{-3t}} \end{aligned}$$

- 8 Solve the differential equation $(x^2 + 1) \frac{dy}{dx} = 6xy$ where $x = 1$ and $y = 2$ giving your answer as y in terms of x .

$$\begin{aligned} (x^2 + 1) \frac{dy}{dx} &= 6xy \\ \int \frac{dy}{y} &= \int \frac{6x}{x^2 + 1} dx \\ \ln |y| &= 3 \int \frac{2x}{x^2 + 1} dx \\ \ln |y| &= 3 \ln |x^2 + 1| + C \\ x=1, y=2 & \quad \ln 2 = 3 \ln 2 + C \\ -2 \ln 2 &= C \\ \therefore \ln y &= 3 \ln (x^2 + 1) - 2 \ln 2 \\ \ln y &= \ln (x^2 + 1)^3 - \ln 2^2 \\ y &= \frac{(x^2 + 1)^3}{4} \end{aligned}$$

9 Given that $y = e^{2x} + e^{-2x}$, determine the values of constants a and b that satisfy the following

4

differential equation $\frac{d^2y}{dx^2} + a\frac{dy}{dx} + by = 5e^{2x} + e^{-2x}$.

$$\begin{aligned}y &= e^{2x} + e^{-2x} \\ \frac{dy}{dx} &= 2e^{2x} - 2e^{-2x} \\ \frac{d^2y}{dx^2} &= 4e^{2x} + 4e^{-2x} \\ \frac{d^2y}{dx^2} + a\frac{dy}{dx} + by &= 4e^{2x} + 4e^{-2x} + 2ae^{2x} - 2ae^{-2x} + be^{2x} + be^{-2x} \\ &= (4 + 2a + b)e^{2x} + (4 - 2a + b)e^{-2x} \\ \text{Equating coefficients:} \quad &4 + 2a + b = 5 \quad 4 - 2a + b = 1 \\ &2a + b = 1 \quad (1) \quad -2a + b = -3 \quad (2) \\ &+ \quad -2a + b = -3 \quad (2) \quad -2a - 1 = -3 \\ &2b = -2 \quad -2a = -2 \\ &b = -1 \quad a = 1 \\ \therefore a &= 1 \quad b = -1\end{aligned}$$